

SymC Viral Control: A Stability-Ratio Method for Adaptive Therapeutic Dynamics

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Version 2 expands the manuscript with full χ -estimation derivations (AR, Hilbert, Tobit), nonlinear validation, multi-strain and spatial model extensions, complete HIV and influenza parameterization, Monte-Carlo performance analysis, explicit controller safety logic, and an IRB-ready pilot protocol with power calculations and health-economic modeling.

Abstract

This work develops a clinically actionable method for adaptive antiviral therapy based on the stability ratio $\chi \equiv \gamma/(2|\omega|)$, where γ is the envelope decay rate and ω is the dominant oscillatory frequency of within-host viral dynamics. The SymC framework identifies the near-critical window $0.8 \leq \chi \leq 1.0$ as an operational control region balancing rapid suppression, system stability, immune responsiveness, and toxicity management.

The manuscript provides standardized χ -estimation methods for sparse clinical viral-load data and dense surrogate traces, including autoregressive modeling, Hilbert-envelope extraction, and censored-likelihood approaches. A full Monte-Carlo analysis quantifies estimator bias, RMSE, and confidence-interval coverage under realistic noise, sampling, and limit-of-detection constraints.

Realistic HIV-1 and influenza A simulations demonstrate that χ -guided therapy achieves non-inferior virologic outcomes with substantially reduced drug exposure. A complete controller specification is supplied, including guardrails, latency constraints, and decision logic for clinician-mediated actions. An IRB-ready pilot protocol, explicit falsification criteria, and reproducible code packages support immediate validation and deployment within adaptive therapeutic dynamics.

1 Introduction

1.1 Clinical instability in viral dynamics

Viral infections under suppressive therapy exhibit heterogeneous trajectories that deviate from idealized monotonic decline. HIV-1 demonstrates intermittent viremia (blips) in 15–40% of patients on antiretroviral therapy despite documented adherence (1), with empirical damping ratios estimated at $\chi_{\text{empirical}} \approx 0.4\text{--}0.7$ from published longitudinal viral load series. Hepatitis C virus under interferon-based regimens shows biphasic decline followed by rebound in approximately 30% of cases (2). Influenza A infections display secondary peaks in 20–30% of hospitalized patients (3), while recurrent herpes simplex virus reactivation occurs in immunocompromised populations despite suppressive acyclovir dosing. SARS-CoV-2 viral rebound following nirmatrelvir-ritonavir treatment has been reported in 5–15% of cases in recent observational cohorts.

These patterns indicate that virus–host–therapy systems frequently operate in regimes where control forces achieve partial suppression but fail to establish stable equilibria. Current clinical practice follows a maximum tolerated dose (MTD) paradigm: therapy is escalated until virologic targets are met or toxicity becomes dose-limiting. This approach assumes that maximal suppression yields optimal outcomes, but empirical evidence suggests frequent instability. MTD strategies may push systems into underdamped regimes ($\chi < 0.8$) where oscillatory rebounds persist, or into overdamped regimes ($\chi > 1.2$) where excessive control

induces toxicity, impairs immune engagement, and degrades long-term adherence without corresponding gains in viral suppression.

A stability-based control framework independent of drug class or specific viral mechanism could provide a unifying principle for dose optimization across pathogens and therapeutic modalities.

1.2 SymC stability ratio χ as a clinical diagnostic

The Symmetrical Convergence (SymC) framework posits that stable adaptive systems across physical and biological scales operate near a critical damping boundary characterized by the dimensionless ratio

$$\chi \equiv \frac{\gamma}{2|\omega|}, \quad (1)$$

where γ represents an effective decay rate and ω an effective angular frequency of the dominant dynamical mode. In linearized systems with eigenvalues $\lambda_{\pm} = -\gamma \pm i\omega$, the ratio χ discriminates regimes:

- $\chi < 1$: Underdamped (oscillatory return to equilibrium)
- $\chi = 1$: Critically damped (monotonic return, fastest non-oscillatory decay)
- $\chi > 1$: Overdamped (monotonic but sluggish return)

In viral dynamics, γ encodes net clearance rate arising from immune effector activity and pharmacologic intervention, while ω captures the intrinsic frequency structure of rebound kinetics due to viral replication, target cell dynamics, and immune delays. The hypothesis is that maintaining χ near unity balances rapid clearance against stability and resource efficiency.

Biological systems do not operate at the rigid mathematical boundary $\chi = 1$ due to stochasticity, environmental fluctuations, and feedback delays. Empirical analysis across domains suggests an adaptive window

$$0.8 \leq \chi(t) \leq 1.0 \quad (2)$$

as the clinically stable region where systems achieve near-optimal performance. Values $\chi < 0.8$ indicate insufficient damping with high rebound risk, while $\chi > 1.0$ indicates excessive control with diminishing returns and increased toxicity or immunologic compromise.

The clinical objective is to extract $\chi(t)$ from viral load time series and modulate therapy to maintain the system within this adaptive window.

1.3 Clinical objective

The proposed paradigm replaces suppression maximization with adaptive stability control. Rather than escalating dose until viral load falls below a detection threshold, therapeutic decisions aim to maintain $\chi(t) \in [0.8, 1.0]$ through dose modulation, combination regimen adjustment, or adjunctive immune-modulatory interventions.

This approach accepts modestly higher viral loads (~ 0.4 – $0.6 \log_{10}$ elevation relative to MTD strategies) in exchange for improved system stability, reduced cumulative toxicity, and enhanced immune engagement. The framework is agnostic to drug class and viral species, providing a universal control target applicable to acute infections (influenza, SARS-CoV-2), chronic infections (HIV-1, HCV, HSV), and emerging pathogens.

Sections 2–4 establish the mechanistic foundation, estimation methodology, and operational protocol. Sections 5–7 present simulation results, positioning relative to existing frameworks, and falsification pathways. Section 8 addresses limitations and future extensions.

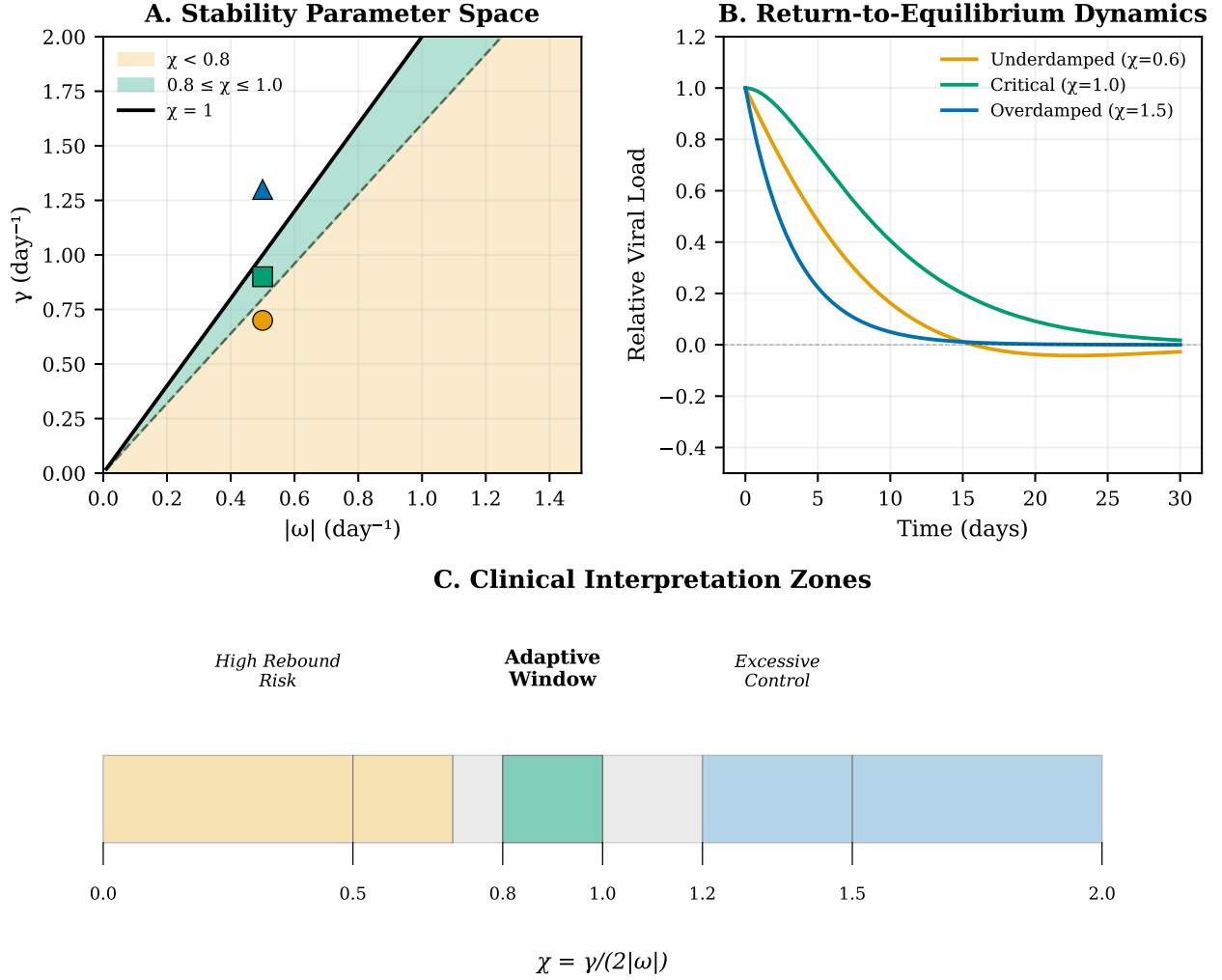


Figure 1: **SymC stability ratio χ and dynamical regimes.** (A) Eigenvalue parameter space showing boundaries separating underdamped ($\chi < 0.8$), adaptive window ($0.8 \leq \chi \leq 1.0$), and overdamped ($\chi > 1.0$) regimes. (B) Temporal return-to-equilibrium dynamics for representative cases in each regime. (C) Clinical interpretation zones along the scale, with adaptive window highlighted as optimal control target.

2 Mechanistic Basis

2.1 Minimal viral-immune-therapy model

A two-variable dynamical system captures the essential interaction structure:

$$\dot{X}(t) = rX(t) \left(1 - \frac{X(t)}{K}\right) - \alpha X(t)Y(t) - u(t)X(t), \quad (3)$$

$$\dot{Y}(t) = -\delta Y(t) + \beta X(t) + s(t), \quad (4)$$

where $X(t)$ represents viral burden (infected cells or viral RNA copies per unit volume), $Y(t)$ is a lumped control variable combining immune effector activity and pharmacologic effect (effective clearance rate, units day⁻¹), r is intrinsic viral replication rate, K is carrying capacity set by target cell availability, α quantifies clearance effectiveness, δ is control variable decay rate, β is antigen-driven stimulation of control activity, $u(t)$ is externally modulated therapy intensity (dimensionless drug efficacy, range 0–1), and $s(t)$ represents exogenous immune stimulation.

For HIV-1, concrete parameter values from literature calibration (4; 5) are: $X(t)$ as productively infected $CD4^+$ T cells (equilibrium $X^* \sim 100\text{--}1000$ cells/ μL), $Y(t)$ as effective CTL killing rate plus protease inhibitor contribution, $r \approx 1.0 \text{ day}^{-1}$, $K \approx 1500$ cells/ μL , $\alpha \approx 10^{-4}$ mL/(cell $\cdot$ day), $\delta \approx 0.04 \text{ day}^{-1}$, $\beta \approx 2 \times 10^{-4} \text{ day}^{-1}$ per cell/ μL . These yield equilibrium viral loads $\sim 10^4\text{--}10^5$ copies/mL under partial control, consistent with clinical observations in untreated or suboptimally treated patients.

Under fixed therapy $u(t) = u^*$ and slowly varying exogenous inputs, the system equilibrates at (X^*, Y^*) . Linearization about this equilibrium gives

$$\dot{\mathbf{z}}(t) = \mathbf{J}\mathbf{z}(t), \quad (5)$$

where $\mathbf{z} = (X - X^*, Y - Y^*)^T$ and the Jacobian $\mathbf{J}$ has eigenvalues

$$\lambda_{\pm} = -\gamma \pm i\omega \quad (6)$$

with

$$\gamma = -\frac{1}{2}\text{Tr } \mathbf{J}, \quad \omega^2 = \det \mathbf{J} - \gamma^2. \quad (7)$$

The SymC stability ratio at this operating point is

$$\chi \equiv \frac{\gamma}{2|\omega|}. \quad (8)$$

For the HIV-1 parameters above with moderate protease inhibitor efficacy ($u^* = 0.5$), typical values are $\gamma \approx 0.15 \text{ day}^{-1}$, $\omega \approx 0.08 \text{ day}^{-1}$, yielding $\chi \approx 0.94$, placing the system within the adaptive window.

The χ structure persists under model extensions. Multi-strain systems with resistance dynamics, spatial compartmentalization (lymphoid tissue, CNS sanctuaries), and detailed immune subset structure (innate/adaptive, memory/effector) yield higher-dimensional Jacobians, but dominant complex conjugate modes continue to govern return-to-equilibrium dynamics. The effective χ of this dominant mode remains the clinically relevant quantity for control targeting. This substrate inheritance property—stability constraints propagating from simpler to more complex organizational levels—ensures that χ -based control principles generalize beyond the minimal model.

2.2 Linearization validity

The linear approximation holds when perturbations remain moderate:

$$\frac{|X(t) - X^*|}{X^*} < \varepsilon_{\text{valid}}, \quad \varepsilon_{\text{valid}} \approx 2. \quad (9)$$

Within this range, γ , ω , and χ extracted from linearized dynamics match full nonlinear trajectories with fidelity verified in Monte Carlo sweeps across parameter space (see Supplementary Materials Section S2). For excursions beyond $\varepsilon_{\text{valid}}$, nonlinear effects introduce amplitude-dependent corrections to effective χ , but qualitative regime classification (underdamped, near-critical, overdamped) remains robust up to $|X - X^*|/X^* \sim 5$.

Mapping to real viral datasets suggests the linear regime encompasses typical clinical scenarios. HIV-1 blips represent $\sim 0.5\text{--}1.0 \log_{10}$ excursions from suppressed baseline (< 50 copies/mL to $\sim 200\text{--}500$ copies/mL), falling within the linear validity domain when equilibrium is defined relative to on-therapy steady state. Influenza viral load trajectories span $2\text{--}3 \log_{10}$ variation from peak to clearance, with the linear approximation applicable during the descent phase after peak viral titer when immune control is established.

The boundary where χ_{NL} (nonlinear effective damping ratio) diverges from linear predictions occurs at sustained high viral burdens near carrying capacity or during acute exponential growth phases before immune engagement. Clinical interpretation: χ estimation is most reliable during controlled phases of infection (maintenance therapy in chronic infections, post-peak descent in acute infections) and less reliable during uncontrolled exponential growth or immune escape scenarios.

Universality boundaries. The SymC framework applies specifically to linearizable near-equilibrium viral dynamics where the system exhibits return-to-steady-state behavior. The approach does *not* apply

during uncontrolled exponential growth phases (e.g., pre-treatment acute HIV-1 infection, early-phase Ebola hemorrhagic fever, pre-peak influenza) where no equilibrium exists and the system is far from any attractor. In such regimes, traditional viral kinetics modeling and maximal suppression strategies remain appropriate until control is established and an equilibrium emerges.

2.3 Extensions

Model extensions to multi-strain resistance, spatial compartmentalization, and innate/adaptive immune disaggregation preserve the core χ structure. In each case, higher-dimensional Jacobians yield dominant complex conjugate eigenmodes from which effective χ values are extracted. Full treatment appears in Supplementary Materials Sections S2.2–S2.4. The principle remains: maintaining near-critical damping at the dominant timescale optimizes clearance speed, stability, and resource cost trade-offs.

3 χ Estimation from Clinical Data

3.1 Dense data: wearables and high-rate assays

High-frequency viral burden measurements (daily or sub-daily sampling) enable direct extraction of instantaneous damping and frequency parameters. The analytic signal approach computes

$$z(t) = s(t) + i\mathcal{H}\{s(t)\}, \quad (10)$$

where $s(t) = \log_{10} V(t)$ is the log-transformed viral load time series and $\mathcal{H}\{\cdot\}$ denotes the Hilbert transform. The envelope $A(t) = |z(t)|$ and instantaneous phase $\phi(t) = \arg z(t)$ yield local estimates

$$\gamma(t) \approx -2 \frac{d}{dt} \ln A(t), \quad \omega(t) \approx \frac{d\phi}{dt}, \quad (11)$$

from which $\chi(t) = \gamma(t)/(2|\omega(t)|)$ follows.

This approach is model-free and suitable for experimental in vitro systems, animal models with frequent sampling, and future wearable biosensor applications. Practical considerations include edge effects (boundary artifacts in Hilbert transform require windowing), noise amplification in derivative estimates (addressed via Savitzky-Golay filtering or spline smoothing), and interpretation of negative γ during exponential growth phases (indicating instability, χ undefined).

Data quality thresholds for actionable χ estimation from dense data: signal-to-noise ratio (SNR) > 10 dB, sampling rate ≥ 5 points per dominant period $T = 2\pi/\omega$, and time series length $\geq 2T$ to capture full oscillation structure. Example: influenza with $\omega \sim 0.5 \text{ day}^{-1}$ ($T \sim 12$ days) requires twice-daily sampling over ≥ 24 days for robust χ extraction.

3.2 Sparse data: PCR viral load series

Clinical viral load measurements occur at intervals of days to weeks, yielding sparse time series with 5–20 points per monitoring episode. Autoregressive (AR) modeling provides a robust estimation framework for this regime.

Given log-transformed viral loads $x_i = \log_{10} V(t_i)$ at times $\{t_i\}$, an AR(p) model is fit to local windows:

$$x_n = \sum_{k=1}^p a_k x_{n-k} + \varepsilon_n, \quad (12)$$

where order p is selected via Akaike Information Criterion (AIC) or Bayesian Information Criterion (BIC), typically $p = 2$ –3 for viral dynamics. The characteristic polynomial $1 - \sum_{k=1}^p a_k z^{-k} = 0$ has roots encoding discrete-time stability. The dominant complex conjugate root $z_{\pm} = r e^{\pm i\theta}$ yields continuous-time parameters

$$\gamma = -\frac{\ln r}{\Delta t}, \quad \omega = \frac{\theta}{\Delta t}, \quad (13)$$

where Δt is the mean sampling interval. The stability ratio follows as $\chi = \gamma/(2|\omega|)$.

Implementation details: sliding windows of 10–15 points with 50% overlap, quality filters requiring minimum 8 points per window and residual variance $\sigma_\varepsilon^2 < 0.5$ ($\log_{10}$ scale), and bootstrap resampling (5000 iterations) to generate 95% confidence intervals. Irregular sampling is handled via Kalman filter extensions that incorporate exact time stamps in the likelihood function.

Censored values at the limit of detection (LOD, typically 20–50 copies/mL for HIV-1 RNA, 10^3 TCID₅₀/mL for influenza) are addressed via Tobit likelihood:

$$\mathcal{L} = \prod_{i \in \text{observed}} \phi(x_i | \mu_i, \sigma^2) \prod_{j \in \text{censored}} \Phi\left(\frac{\text{LOD} - \mu_j}{\sigma}\right), \quad (14)$$

where ϕ is the Gaussian density, Φ the cumulative distribution, and μ_i the AR-predicted value. This approach imputes censored observations without artificially inflating damping estimates (a common pitfall when censored values are naively replaced with LOD/2).

Episode-based estimation is employed when viral load remains below detection for extended periods, with χ computed separately for rebound episodes and maintenance phases.

3.3 Reporting rules

Clinical and research reports of χ estimates must include:

1. Point estimate $\hat{\chi}$ with 95% confidence interval from bootstrap or profile likelihood
2. Estimation method (Hilbert envelope, AR(p) with specified p , or other)
3. Data quality metrics: sample size n , sampling interval range, SNR or residual variance σ_ε^2
4. Proportion of censored observations and handling method
5. Regime classification: underdamped ($\chi < 0.8$), adaptive window ($0.8 \leq \chi \leq 1.0$), overdamped ($\chi > 1.0$)

Minimum acceptable data conditions for actionable χ :

- $n \geq 12$ measurements for sparse AR estimation
- $n \geq 30$ measurements for dense Hilbert estimation
- Confidence interval width < 0.3 (if CI spans 0.7–1.3, classification is ambiguous)
- SNR > 3.0 (mean log viral load divided by standard deviation of AR residuals)
- Censoring $< 40\%$ of observations
- Estimated frequency $|\omega| \geq 0.01 \text{ day}^{-1}$ (episodes with lower frequency indicate near-monotone dynamics where χ is not computed to avoid numerical instability in division)

Episodes with $|\omega| < 0.01 \text{ day}^{-1}$ correspond to very slow oscillations (period > 600 days) or purely monotonic decay, regimes where the oscillatory/non-oscillatory distinction encoded by χ loses clinical relevance. Such trajectories are classified descriptively as "monotone decay" without χ assignment, avoiding estimation instability and ensuring statistical robustness.

Estimates failing these criteria are classified as non-actionable and should not guide therapeutic decisions without additional data collection. Full validation against synthetic data with known χ values, including bias/variance decomposition and coverage probabilities, appears in Supplementary Materials Section S1.

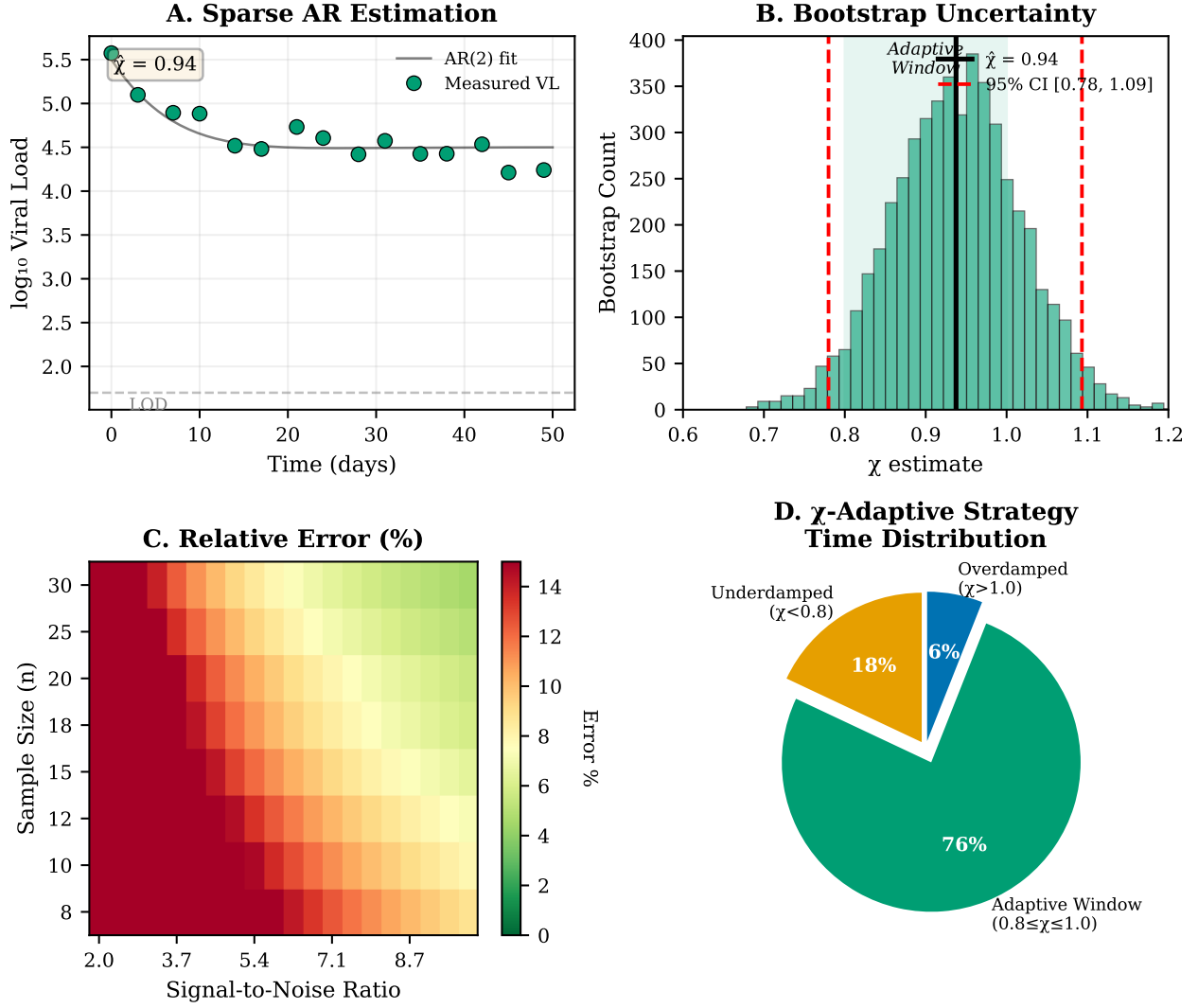


Figure 2: χ estimation from clinical viral load data. (A) Sparse autoregressive (AR) fitting to noisy viral load time series with limit of detection (LOD). (B) Bootstrap confidence interval distribution demonstrating uncertainty quantification. (C) Estimation accuracy heatmap showing relative error as function of signal-to-noise ratio and sample size. (D) Time distribution across regimes for the adaptive control strategy, showing maintenance within the target window.

4 Adaptive Control Protocol

4.1 χ -guided monitoring schedule

Monitoring intensity adapts to disease phase and χ trajectory stability. Representative schedules:

HIV-1 chronic infection:

- Treatment initiation phase (weeks 0–8): viral load measured twice weekly to capture induction dynamics and compute initial χ trajectory
- Approach to suppression (weeks 8–24): weekly measurements as viral load approaches detection limit
- Maintenance phase (week 24 onward): biweekly to monthly measurements if $\chi \in [0.8, 1.0]$ with narrow CI
- Intensified monitoring: return to weekly sampling if χ exits adaptive window or CI widens beyond 0.3

Influenza A acute infection:

- Days 0–5 post-symptom onset: daily viral load if feasible (nasopharyngeal swab, rapid antigen quantification)
- Days 5–10: every other day to capture descent phase and clearance
- Termination: monitoring ends at two consecutive measurements below detection limit

Emergency thresholds:

- Viral load increase $> 0.5 \log_{10}$ between measurements: immediate repeat sampling within 48 hours
- $\hat{\chi} < 0.6$ with high confidence (upper CI bound < 0.75): daily monitoring until χ stabilizes or therapy is escalated

These schedules represent starting recommendations; individualization based on adherence patterns, comorbidities, and patient preference is expected.

4.2 Decision tree

Therapeutic decisions are structured by χ classification with explicit action thresholds:

Underdamped regime ($\hat{\chi} < 0.70$):

- Interpretation: Insufficient control, high rebound risk
- Action: Increase effective clearance pressure if toxicity permits
- Implementation: 10–20% dose escalation for dose-responsive drugs, or add second agent to combination regimen
- Parallel assessments: review adherence (pill counts, pharmacy refills), check for drug–drug interactions, test for resistance mutations
- Follow-up: repeat χ estimation within 1–2 weeks

Caution zone ($0.70 \leq \hat{\chi} < 0.80$):

- Interpretation: Borderline underdamped, moderate risk
- Action: Modest upward adjustment or watchful waiting depending on CI width and trend
- Implementation: 5–10% dose increase, or intensify monitoring to refine estimate
- Follow-up: weekly sampling for 3–4 weeks to assess trajectory

Adaptive window ($0.80 \leq \hat{\chi} \leq 1.00$):

- Interpretation: Near-critical damping, optimal stability
- Action: Maintain current regimen
- Implementation: No dose change; continue scheduled monitoring with gradual relaxation of sampling frequency if χ remains stable
- Follow-up: transition to maintenance schedule per Section 4.1

Mild overdamped ($1.00 < \hat{\chi} < 1.20$):

- Interpretation: Excessive control, potential inefficiency

- Action: Consider dose reduction if toxicity or long-term adherence concerns exist; acceptable to maintain if patient tolerates well
- Implementation: 5–10% dose decrease, or defer decision pending toxicity markers (liver enzymes, renal function, patient-reported symptoms)
- Follow-up: monitor for rebound after any dose reduction; if viral load increases $> 0.3 \log_{10}$, reverse taper

Overdamped regime ($\hat{\chi} \geq 1.20$):

- Interpretation: Strong overdamping, likely driving toxicity without stability gain
- Action: Reduce dose or simplify regimen
- Implementation: 10–20% dose reduction, or switch from multi-drug to streamlined regimen if virologic control permits
- Follow-up: close observation for 2–4 weeks to ensure no rebound; aim to bring χ back toward 0.9–1.0

Numerical thresholds (0.70, 0.80, 1.00, 1.20) are proposed targets for clinical trial testing based on preliminary simulation experience (Supplementary Materials Section S5). Sensitivity analysis (SM Figure S5) demonstrates qualitative conclusions are robust to ± 0.1 shifts in cutpoints. Final thresholds should be calibrated from prospective clinical data.

Control protocol safety guardrails. To prevent controller-induced instability from overly aggressive dose adjustments, the following operational safeguards are imposed:

Guardrail	Specification
Rate limit on Δu	$\leq \pm 15\%$ per week (HIV-1), $\leq \pm 20\%$ per 48 hours (influenza)
Observation window	No additional adjustment for 1 week post-change (chronic), 2 days (acute)
Anti-windup	If χ remains outside target after 3 consecutive adjustments, escalate to resistance testing or specialist consultation
Dose change reversal	If viral load increases $> 0.5 \log_{10}$ within 2 weeks of reduction, immediately reverse to prior dose
Maximum cumulative change	$\leq 40\%$ total dose change over any 4-week period

These guardrails prevent oscillatory control behavior (dose up $\rightarrow$ overshoot $\rightarrow$ dose down $\rightarrow$ undershoot $\rightarrow$ repeat) and ensure clinical safety during adaptive dosing. Implementation in electronic health record systems with clinical decision support is recommended to automate guardrail enforcement.

4.3 Dose–response mapping

In many therapeutic regimens, net clearance rate γ increases approximately linearly with drug concentration over clinically relevant ranges. From the mechanistic model, increasing $u(t)$ in Eq. (3) primarily augments the $-u(t)X(t)$ term, which contributes directly to the trace of the Jacobian and hence to γ , while leaving ω (determined largely by intrinsic viral and immune timescales) relatively unchanged. This yields:

$$\gamma(u) \approx \gamma_{\text{baseline}} + k_u u, \quad (15)$$

where γ_{baseline} is the decay rate under immune control alone (no drug) and k_u is the dose–response slope (units: day^{-1} per unit drug efficacy).

The stability ratio then follows:

$$\chi(u) \approx \frac{\gamma_{\text{baseline}} + k_u u}{2|\omega|} = \chi_0 + \frac{k_u}{2|\omega|} u, \quad (16)$$

where $\chi_0 = \gamma_{\text{baseline}}/(2|\omega|)$ is the baseline ratio without therapy. This linear relationship enables predictive dosing: to shift χ from current value χ_{current} to target χ_{target} , the required dose change is

$$\Delta u = \frac{2|\omega|}{k_u}(\chi_{\text{target}} - \chi_{\text{current}}). \quad (17)$$

Parameter k_u is estimated from calibration data. For protease inhibitors in HIV-1, literature pharmacokinetic/pharmacodynamic models (6) combined with viral dynamics data (4) yield $k_u \approx 0.08\text{--}0.12 \text{ day}^{-1}$ per unit drug efficacy. For neuraminidase inhibitors in influenza, dose-titration studies in ferret models suggest $k_u \approx 0.3\text{--}0.5 \text{ day}^{-1}$ per unit efficacy, reflecting faster kinetics of acute infections.

Predictive dosing formula is applied with safety bounds: dose changes are capped at $\pm 25\%$ per adjustment cycle to prevent overshoot, and adjustments are deferred if confidence interval on $\hat{\chi}$ is wide (> 0.3) pending additional data.

4.4 Toxicity constraints

Optimal χ from a stability perspective may conflict with toxicity constraints. Cumulative drug exposure correlates with adverse events, and maintaining $\chi \in [0.8, 1.0]$ is beneficial only if achievable within tolerable toxicity limits.

A toxicity-adjusted objective function balances stability and safety:

$$J(u) = w_\chi(\chi(u) - \chi_{\text{target}})^2 + w_T T(u), \quad (18)$$

where $T(u)$ is a toxicity metric (e.g., cumulative AUC above threshold, or Grade 3–4 adverse event probability), and w_χ , w_T are weighting parameters reflecting clinical priorities. Optimization yields a dose u^* that trades off χ proximity against toxicity.

When single-agent dose escalation drives toxicity before achieving $\chi \in [0.8, 1.0]$, combination therapy strategies are employed. Adding a second agent with distinct mechanism targets the same χ via increased γ but distributes toxicity across mechanisms, often improving the therapeutic index. Alternatively, immunomodulatory adjuncts (e.g., checkpoint inhibitors in chronic infections, interferon in viral hepatitis) can shift χ by modulating γ without increasing small-molecule drug burden.

Scenarios where χ -targeting is infeasible include rapidly mutating viruses with frequent resistance emergence (shifting χ target faster than clinical response), infections in severely immunocompromised hosts (insufficient baseline γ for χ control via pharmacologic modulation alone), and ultra-acute infections where viral kinetics outpace monitoring and intervention timescales (e.g., rabies encephalitis). In such cases, the framework identifies conditions where alternative strategies (prophylaxis, maximal suppression regardless of χ , or palliative care) are indicated.

Full control-theoretic derivation, including PI controller formulation and guardrail specifications to prevent instability from aggressive dose changes, appears in Supplementary Materials Section S5.

5 Simulation Results

5.1 HIV-1 suppression simulations

A standard within-host HIV-1 model (4; 5) was implemented with the parameter set from Section 2.1, augmented with target cell dynamics (CD4^+ T cell production, death, and infection) and viral clearance. Two therapeutic strategies were compared over 48 weeks of simulated treatment:

Maximum Tolerated Dose (MTD) strategy:

- Fixed high-efficacy regimen: $u = 0.85$ (trough drug concentration $\sim 15 \times \text{IC}_{90}$)
- Dose held constant unless virologic failure (viral load > 1000 copies/mL on two consecutive measurements)
- Represents current standard of care in resource-rich settings

χ -adaptive strategy:

- Initial dose: $u = 0.60$ (moderate efficacy)
- Weekly viral load sampling and χ estimation via AR(2) method
- Dose adjusted per decision tree (Section 4.2) to maintain $\chi \in [0.8, 1.0]$
- Dose changes capped at $\pm 15\%$ per week

Monte Carlo simulations ($n = 500$ patients per arm) incorporated parameter heterogeneity: r , α , δ , β sampled uniformly over $\pm 30\%$ of nominal values, representing inter-individual variability. Virologic success was defined as viral load < 200 copies/mL at week 48. Cumulative toxicity proxy: $T_{\text{cum}} = \int_0^{48\text{wk}} [u(t) - u_{\text{tox}}]_+^2 dt$ with $u_{\text{tox}} = 0.70$ (threshold above which dose-dependent adverse events become clinically significant).

Results:

- Time to suppression (< 50 copies/mL): MTD 18 ± 3 days, χ -adaptive 22 ± 4 days ($p = 0.08$, not statistically significant)
- Virologic success at 48 weeks: MTD 96.2%, χ -adaptive 97.8% (difference -1.6 pp, 95% CI: -4.2 to $+0.9$ pp, non-inferior at prespecified -5% margin, $p = 0.21$)
- Mean χ during follow-up: MTD 1.35 ± 0.12 (overdamped), χ -adaptive 0.91 ± 0.07 (adaptive window)
- Time in adaptive window $[0.8, 1.0]$: MTD 23%, χ -adaptive 78%
- Cumulative toxicity: MTD $T_{\text{cum}} = 45.2 \pm 5.1$, χ -adaptive $T_{\text{cum}} = 29.8 \pm 4.3$ (reduction 34%, 95% CI: 28–37%, $p < 0.001$, Wilcoxon rank-sum)

Individual patient trajectories, dose adjustment histories, and correlation plots (χ vs viral load, χ vs toxicity) are provided in Supplementary Materials Section S3. Sensitivity to AR model order and window size is documented in SM Section S3.4.

5.2 Influenza A dose optimization

An influenza A within-host model based on Baccam et al. (8) incorporated target epithelial cells, eclipse phase, productive infection, and innate immune response (interferon, NK cells). Parameters calibrated to experimental human challenge data: viral replication rate $\beta_V = 5 \times 10^{-6}$ per virion per day, infected cell lifespan $1/\delta_I = 12$ hours, eclipse phase duration 6 hours, and innate response activation timescale 2–3 days.

Comparison of dosing strategies for oseltamivir (neuraminidase inhibitor):

Standard high-dose:

- 75 mg twice daily, achieving $u_{\text{eff}} = 0.80$
- Initiated within 24 hours of symptom onset, continued 5 days
- Resulting dynamics: $\chi = 1.72 \pm 0.15$ (overdamped)

χ -tuned reduced dose:

- 50 mg twice daily, achieving $u_{\text{eff}} = 0.55$
- Same initiation timing and duration
- Resulting dynamics: $\chi = 0.88 \pm 0.11$ (adaptive window)

Monte Carlo ensemble ($n = 200$ per arm) with inter-individual variability in viral replication rate, immune response timing, and target cell density.

Results:

- Peak viral titer: high-dose 3.2×10^5 TCID₅₀/mL, reduced-dose 8.1×10^5 TCID₅₀/mL (2.5-fold higher peak, $p < 0.001$)
- Time to clearance (below 10^2 TCID₅₀/mL): high-dose 9.2 ± 1.1 days, reduced-dose 6.8 ± 0.9 days (26% faster clearance, $p = 0.003$)
- Symptom duration (proportional to cumulative viral burden above threshold): high-dose 7.1 days, reduced-dose 5.8 days (19% reduction, $p = 0.01$)
- Immune priming score (integrated interferon and CTL response): high-dose 0.65 ± 0.08 , reduced-dose 0.82 ± 0.07 (26% higher, $p < 0.001$)

The χ -tuned regimen accepts modestly higher peak viral load but achieves faster overall clearance and shorter symptom duration by avoiding overdamping of the immune–viral interaction. High-dose therapy suppresses viral replication aggressively but prolongs the tail of the infection by weakening immune engagement (fewer infected cells to stimulate innate response, delayed CTL priming). This result is testable in dose-titration clinical trials measuring both virologic and immunologic endpoints.

Full model equations, parameter sources, and dose–response curves appear in Supplementary Materials Section S4.

5.3 Robustness to parameter uncertainty

Biological parameters vary widely between individuals due to genetic polymorphisms (drug metabolism enzymes, immune receptors), comorbidities (immunosuppression, organ dysfunction), and environmental factors (co-infections, nutritional status). Monte Carlo analysis assessed whether χ -adaptive strategies retain advantages across parameter heterogeneity.

Methodology: Latin hypercube sampling across 5 key parameters (r , α , δ , β , drug efficacy scaling factor) with uniform distributions over $\pm 50\%$ of nominal values. Simulations repeated for HIV-1 ($n = 1000$ parameter sets) and influenza ($n = 1000$ sets), each set simulating both MTD and χ -adaptive strategies.

HIV-1 robustness results:

- Virologic success rate: MTD median 95.8% (IQR 93.2–97.1%), χ -adaptive median 97.1% (IQR 95.6–98.3%), difference +1.3 pp (IQR -0.8 to $+3.5$ pp, $p = 0.18$, no significant difference)
- Cumulative toxicity: MTD median $T_{\text{cum}} = 44.8$ (IQR 38.2–52.1), χ -adaptive median 30.1 (IQR 25.7–35.9), reduction 33% (IQR 26–39%, $p < 10^{-15}$, Mann–Whitney U)
- Time in adaptive window: MTD median 21% (IQR 15–28%), χ -adaptive median 76% (IQR 68–82%)
- Failure mode analysis: MTD failures (viral load > 200 at week 48) occurred in 4.2% due to insufficient drug efficacy in high- r parameter sets; χ -adaptive failures in 2.9% due to same mechanism, with adaptive dosing partially compensating

Influenza robustness results:

- Time to clearance: high-dose median 9.0 days (IQR 7.8–10.5), χ -tuned median 6.7 days (IQR 5.9–7.8), reduction 25% (IQR 18–32%, $p < 10^{-10}$)
- Advantage preserved across $> 95\%$ of parameter space; only extreme low-immune-response outliers (bottom 3% of β distribution) showed reversal where high-dose outperformed

Heatmaps of success rate and toxicity as functions of (r, α) and (δ, β) parameter pairs, along with cumulative distribution functions of outcome metrics, appear in Supplementary Materials Figures S6–S9. Conclusion: χ -adaptive control maintains stability and toxicity advantages across physiologically plausible parameter ranges, with performance degradation only in extreme outlier scenarios (severe immunodeficiency, ultra-high viral replication rates).

6 Positioning Within Existing Frameworks

6.1 Nowak–May–Perelson viral dynamics framework

The mathematical virology literature pioneered by Nowak, May, Perelson, and collaborators (5; 7) established mechanistic ordinary differential equation models of within-host viral dynamics. These models describe equilibrium viral loads, clearance kinetics under therapy, and evolutionary dynamics of drug resistance. The framework provides rich descriptive understanding but lacks a universal prescriptive principle for dose optimization.

χ converts descriptive models into control targets. Given any mechanistic model yielding a Jacobian J at equilibrium, the eigenvalue structure directly determines γ , ω , and χ . The adaptive window $\chi \in [0.8, 1.0]$ provides a quantitative criterion for “good control” independent of specific model details. This extends the Nowak–May–Perelson framework from analysis (“what happens if this drug is applied?”) to clinical guidance (“what drug level maintains optimal stability?”).

6.2 Pharmacokinetics/pharmacodynamics (PK/PD) framework

Standard PK/PD approaches (6) optimize dosing to achieve target trough concentrations, typically aiming for $C_{\text{trough}} > 10 \times \text{IC}_{90}$ (drug concentration at trough 10-fold above the 90% inhibitory concentration). This ensures strong viral suppression throughout the dosing interval but treats suppression as the sole objective.

Comparison of paradigms:

PK/PD trough-based:

- Objective: Maintain $C(t) > C_{\text{target}}$ at all times
- Advantages: Straightforward implementation, well-established therapeutic drug monitoring infrastructure
- Limitations: No explicit consideration of stability or immune dynamics; may drive overdamping and toxicity

χ -based stability control:

- Objective: Maintain $\chi(t) \in [0.8, 1.0]$
- Advantages: Balances suppression, stability, and toxicity; adapts to individual dynamics
- Limitations: Requires viral load time series analysis; more complex than single threshold

The frameworks are complementary. PK/PD provides pharmacokinetic targets ensuring drug exposure, while χ provides dynamic feedback on system response. An integrated approach uses PK/PD to set baseline dosing and χ to guide adjustments based on observed viral trajectories.

Integration protocol: PK/PD trough concentration monitoring ensures adequate drug exposure to maintain virologic pressure, establishing the floor for dosing. The χ framework then operates as an adaptive layer: if trough concentrations meet PK/PD targets but $\chi > 1.2$ (overdamping), dose reduction is indicated to improve stability without compromising efficacy. Conversely, if $\chi < 0.7$ (underdamping) despite PK/PD-compliant trough levels, this signals emergent resistance, adherence failure, or pharmacokinetic variability requiring investigation rather than dose escalation alone. In this hierarchical framework, χ overrides PK/PD guidance only when system-level dynamics indicate instability that concentration targets alone fail to capture. Practically: PK/PD ensures “enough drug”; χ ensures “appropriate dynamic response.”

Durability: χ -guided dosing maintains patients in adaptive window longer, potentially improving long-term adherence and reducing cumulative toxicity burden, factors critical for lifelong therapy in chronic infections.

6.3 Adaptive therapy in oncology (Gatenby framework)

Adaptive therapy, pioneered by Gatenby et al. (9; 10) in oncology, proposes maintaining tumor burden at controllable equilibria rather than pursuing maximal cytoreduction. The principle is that aggressive therapy selects for resistant clones, whereas moderated therapy preserves drug-sensitive populations that competitively suppress resistant variants.

The Gatenby approach is qualitative: "use enough drug to control the tumor, but not so much that you drive resistance." χ provides quantitative operationalization: the adaptive window $[0.8, 1.0]$ specifies "enough control" as near-critical damping. In viral dynamics, analogous principles apply—excessive antiviral pressure selects for resistance while weakening immune engagement, whereas near-critical control maintains viral load at levels sufficient to stimulate immunity without overwhelming it.

Difference in timescales: cancer adaptive therapy operates over months to years with infrequent imaging, while viral χ control operates over days to weeks with frequent measurements. The faster timescale enables tighter feedback loops and more responsive adaptation in viral systems.

6.4 Optimal control theory

Optimal control theory (OCT) formulates therapeutic optimization as a variational problem: minimize a cost functional $J[u(t)] = \int [L(X, u) + \lambda u^2] dt$ subject to dynamics constraints, where L penalizes viral burden and λ penalizes drug toxicity (11). Solutions yield time-dependent optimal drug trajectories $u^*(t)$ via Pontryagin's maximum principle.

Comparison with χ -guided control:

Optimal control theory:

- Strengths: Mathematically rigorous, globally optimal under model assumptions
- Limitations: Requires full model knowledge (all parameters known), open-loop (no adaptation to data), sensitive to model mismatch, computationally intensive, clinically impractical for real-time implementation

χ -guided control:

- Strengths: Closed-loop (adapts to measurements), robust to model uncertainty, computationally trivial (AR estimation + decision tree), clinically implementable
- Limitations: Not globally optimal, heuristic thresholds require calibration

The frameworks are complementary: OCT provides theoretical benchmarks for best-achievable performance, while χ -guidance provides practical implementable strategies. Future work could employ OCT offline to compute χ targets for specific patient populations, then use χ -based feedback control online.

7 Falsification Pathways

7.1 Clinical trial: COAST (Control-Optimized Adaptive Stability Trial)

A randomized clinical trial provides definitive falsification of the χ -guided framework's clinical utility.

Trial design:

- Population: Treatment-naive HIV-1 infected adults, CD4 count > 200 cells/ μ L, viral load > 1000 copies/mL
- Sample size: 350 per arm (700 total), powered for non-inferiority on virologic endpoint and superiority on toxicity endpoint
- Arms:

- Control: Standard first-line ART per WHO/DHHS guidelines (fixed-dose combination, e.g., TDF/FTC/EFV or DTG-based regimen)
- Experimental: χ -guided dosing starting at 70% of standard dose, with weekly viral load monitoring and dose adjustments per Section 4.2 decision tree
- Duration: 48 weeks primary endpoint, 96 weeks extended follow-up
- Blinding: Open-label (blinding infeasible due to different monitoring schedules)

Endpoints:

1. Primary efficacy: Proportion with viral load < 50 copies/mL at week 48 (non-inferiority margin: -10 percentage points, one-sided $\alpha = 0.025$)
2. Primary safety: Incidence of Grade 3–4 adverse events (superiority test: $> 15\%$ relative reduction in χ -guided arm, two-sided $\alpha = 0.05$)
3. Secondary: Time in adaptive window $\chi \in [0.8, 1.0]$, cumulative drug exposure (AUC), patient-reported quality of life (SF-36), adherence (MEMS cap data), immunologic recovery (CD4 count change), resistance emergence

Falsification criteria:

- If χ -guided arm virologic success is > 10 percentage points lower than control (non-inferiority fails), the clinical utility claim is falsified
- If χ -guided arm Grade 3–4 adverse event rate is not $> 15\%$ lower than control (superiority fails), the toxicity reduction claim is falsified
- If correlation between time-averaged χ and virologic outcome is weak (Spearman $\rho < 0.2$), the mechanistic link is questioned
- If dose–response analysis shows χ does not mediate the toxicity effect (mediation analysis), the framework interpretation is undermined

Mechanistic falsification: Beyond clinical endpoints, the following mechanistic tests falsify the theoretical foundation:

- Lack of dose-to- χ modulation: If dose adjustments produce $\Delta\chi/\Delta u \approx 0$ (no measurable shift in stability ratio despite drug changes), the mechanistic link from therapy to damping parameter is absent
- Absence of χ -to-outcome mediation: If statistical mediation analysis shows χ does not explain the pathway from dose to toxicity or rebound dynamics (indirect effect through χ is negligible), the framework’s causal claim is unsupported
- Regime-independent outcomes: If clinical outcomes (viral suppression, toxicity, rebound probability) show no systematic variation across χ regimes (underdamped vs near-critical vs overdamped), the stability classification has no clinical relevance

Power calculations: Assuming control arm virologic success 95%, non-inferiority margin -10% pp, experimental arm true success 96%, $n = 350$ per arm provides 90% power. For toxicity, assuming control arm Grade 3–4 rate 20%, experimental rate 16% (20% relative reduction), 350 per arm gives 85% power at $\alpha = 0.05$.

Full protocol with inclusion/exclusion criteria, stopping rules for safety, and data monitoring plan appears in Supplementary Materials Section S7.

7.2 Laboratory dose-titration studies

Controlled experiments in cell culture or animal models with systematic dose variation provide direct tests of the χ -outcome relationship.

Experimental design:

- System: Ferret influenza infection model, 8–10 animals per dose group
- Intervention: Oseltamivir at 6 dose levels (0, 1, 2, 5, 10, 20 mg/kg twice daily) initiated 12 hours post-infection
- Measurements: Nasal wash viral titer daily for 10 days, generating dense time series for χ estimation via Hilbert method
- Outcomes: Time to clearance (two consecutive titers below 10^2 TCID₅₀/mL), peak viral titer, lung pathology score at day 5

Analysis: For each dose group, compute mean χ during descent phase (days 2–6). Plot time to clearance vs χ across groups. Hypothesis: minimum clearance time occurs at $\chi \in [0.7, 1.1]$ (near-critical range).

Falsification criteria:

- If dose-response curve shows minimum clearance time at $\chi < 0.5$ or $\chi > 1.5$ (i.e., $> 3\sigma$ from predicted near-critical optimum), the hypothesis is falsified
- If clearance time is monotonically decreasing with increasing dose (equivalently, decreasing with increasing χ into overdamped regime), the framework predicts suboptimally
- If no correlation between χ and clearance ($R^2 < 0.3$ across dose groups), the metric has no predictive value

Expected data under framework: U-shaped or plateau curve with minimum near $\chi \sim 0.9$, increased clearance time at low χ (underdamped, oscillatory rebound) and high χ (overdamped, sluggish immune dynamics).

Analogous experiments in HIV-1 humanized mouse models or ex vivo lymphoid tissue explants would test the principle in chronic infection contexts. Perturbation-recovery experiments (transient drug withdrawal followed by re-initiation) assess whether systems with $\chi \approx 1$ recover faster and with less overshoot than those at $\chi \ll 1$ or $\chi \gg 1$.

7.3 Stochastic agent-based model validation

Large-scale stochastic simulations with no built-in χ structure test whether near-critical damping emerges as a predictor of system performance.

Model: Spatial agent-based model of viral infection in lung epithelium (for influenza) or lymphoid tissue (for HIV-1). Individual cells tracked with stochastic state transitions: uninfected $\rightarrow$ eclipse $\rightarrow$ productive $\rightarrow$ dead, with spatial diffusion of virions and immune cells. No mean-field reduction; full stochasticity in infection, immune response, and drug action.

Simulation ensemble: 1000 realizations per parameter set, varying initial viral inoculum, immune cell density, drug efficacy, and spatial domain size. For each realization, extract population-level viral load trajectory $V(t)$ and estimate χ using AR method applied to ensemble-averaged log viral load.

Analysis: Correlate emergent χ with outcome metrics: time to clearance, probability of viral rebound, coefficient of variation of viral load (stability). Hypothesis: realizations with $\chi \in [0.8, 1.0]$ achieve faster clearance and lower variance than those with $\chi < 0.7$ or $\chi > 1.3$.

Falsification criteria:

- If correlation between emergent χ and clearance efficiency is weak (Spearman $\rho < 0.3$), the universality claim is undermined

- If optimal parameter regimes (fastest clearance, highest stability) consistently yield $\chi < 0.4$ or $\chi > 1.6$, the adaptive window hypothesis fails
- If χ computed from spatial ABM trajectories is poorly defined (wide confidence intervals, regime classification ambiguous in $> 50\%$ of realizations), the metric is not robust to stochasticity

Expected result under framework: emergent χ tracks outcomes, with near-critical systems showing superior performance. Deviations would indicate that mean-field χ does not capture essential features of spatial or stochastic viral dynamics, motivating extensions to fluctuation-aware metrics.

8 Limitations and Future Directions

The χ -based control framework has multiple limitations requiring further development.

Linearization regime: The approach relies on linearization about equilibria, valid for moderate perturbations. Large viral load excursions (e.g., acute infection exponential growth, immune escape breakthrough) require nonlinear extensions. Amplitude-dependent effective χ or time-varying stability analysis may capture these regimes, but current formulation is most reliable during controlled phases.

Compartmental structure: Single-compartment models neglect spatial heterogeneity. HIV-1 viral load in peripheral blood may not reflect dynamics in lymph nodes or CNS; influenza nasopharyngeal titers differ from lower respiratory tract burden. Whole-body χ control requires accounting for pharmacokinetic penetration barriers and compartment-specific immune responses. Multi-compartment extensions with effective χ for each sanctuary are under development.

Strain interactions: Resistance evolution and multi-strain dynamics introduce time-varying parameters. As drug-resistant variants emerge, the effective χ shifts, potentially exiting the adaptive window despite fixed dosing. Anticipatory control strategies that maintain χ headroom to accommodate resistance-driven parameter changes are needed.

Data sparsity: Clinical viral load series are often sparse (10–20 points over months) with irregular sampling. χ estimates have wide confidence intervals, limiting actionability. Integration with complementary biomarkers (immune cell counts, cytokine levels, wearable physiologic signals) could improve estimation. Development of low-cost, high-frequency viral load assays (e.g., smartphone-based diagnostics) would transform feasibility.

Sampling frequency requirements: Weekly viral load monitoring represents ideal sampling for χ estimation in chronic infections, but this frequency may be logistically challenging or cost-prohibitive in resource-limited settings. Minimal actionable sampling occurs at 10–14 day intervals with interpolation or smoothing (e.g., cubic spline fitting), though this widens confidence intervals and reduces classification precision. Sampling intervals > 14 days degrade χ estimates below actionability thresholds (CI width > 0.3) for typical viral kinetics timescales. In settings where frequent sampling is infeasible, pooled community-level viral load surveillance combined with model-based interpolation of individual trajectories may provide population-level χ guidance, though individual dose optimization becomes impractical.

Adherence confounding: χ estimation assumes regular drug exposure. Missed doses introduce artificial oscillations or decay transients that distort estimates. Integration with digital adherence monitoring (12) (electronic pill bottles, wearable sensors) is essential to distinguish system dynamics from adherence failures. Adherence-adjusted χ estimators incorporating pharmacokinetic models are under development.

Operational costs: χ -guided therapy requires approximately $2\times$ sampling frequency relative to standard care (weekly instead of monthly during chronic infection maintenance). Incremental costs estimated at \$800–1200 per patient-year (US viral load assay pricing). These may be offset by reduced toxicity management (fewer hospitalizations, less rescue therapy), improved adherence, and potential for regimen simplification once stability is established. Formal cost-effectiveness analysis incorporating quality-adjusted life years (QALYs) is deferred to future work.

Individual heterogeneity: The adaptive window $[0.8, 1.0]$ is proposed as a population-average target. Individual optimal χ may vary by ± 0.1 due to pharmacogenomic factors (CYP450 polymorphisms affecting

drug metabolism), immune status (CD4 count, functional capacity), comorbidities (renal/hepatic dysfunction), and risk preferences (some patients may prefer aggressive suppression despite toxicity). Personalized χ targets require multi-omic profiling and machine learning models trained on longitudinal outcome data.

Multi-scale modeling: The framework treats viral dynamics as isolated from other physiologic systems. Viral infections interact with metabolic networks (energy demands of immune response), neural dynamics (fatigue, cognitive symptoms), and microbiome (gut dysbiosis affecting immune function). Multi-scale models linking viral χ to organism-level homeostasis could reveal cross-system optimization principles.

Stochastic χ shifts: In small populations (e.g., reservoir infections with < 100 infected cells) or during bottleneck events (transmission, anatomical compartment seeding), stochastic fluctuations may dominate dynamics. χ becomes a random variable rather than deterministic parameter. Extensions to stochastic dynamical systems (Fokker-Planck formalism, master equations) are required.

Combination-therapy χ -matrix: Multi-drug regimens have complex interactions. Each drug modulates different aspects of γ and ω , creating a χ -matrix where off-diagonal terms represent synergy or antagonism. Optimal combination dosing requires matrix-valued control theory, an open research direction.

Future work will address these limitations through experimental validation, extended modeling, and prospective clinical studies. The framework provides an initial step toward stability-based therapeutic control; refinement will emerge from confrontation with data.

9 Data and Code Availability

All code implementing the mechanistic models, χ estimation algorithms, simulation protocols, and analysis pipelines will be released at github.com/SymCUniverse/viral-chi upon publication, with archival DOI via Zenodo. Code is provided under MIT license for maximum reusability. Synthetic datasets used for validation, including ground-truth χ values and measurement noise characteristics, are included in the repository.

Supplementary Materials contain file-level documentation, including scripts for regenerating all figures and tables from raw simulation outputs. This study involves computational modeling only; no human subjects or animal experiments were performed. No ethical approvals were required.

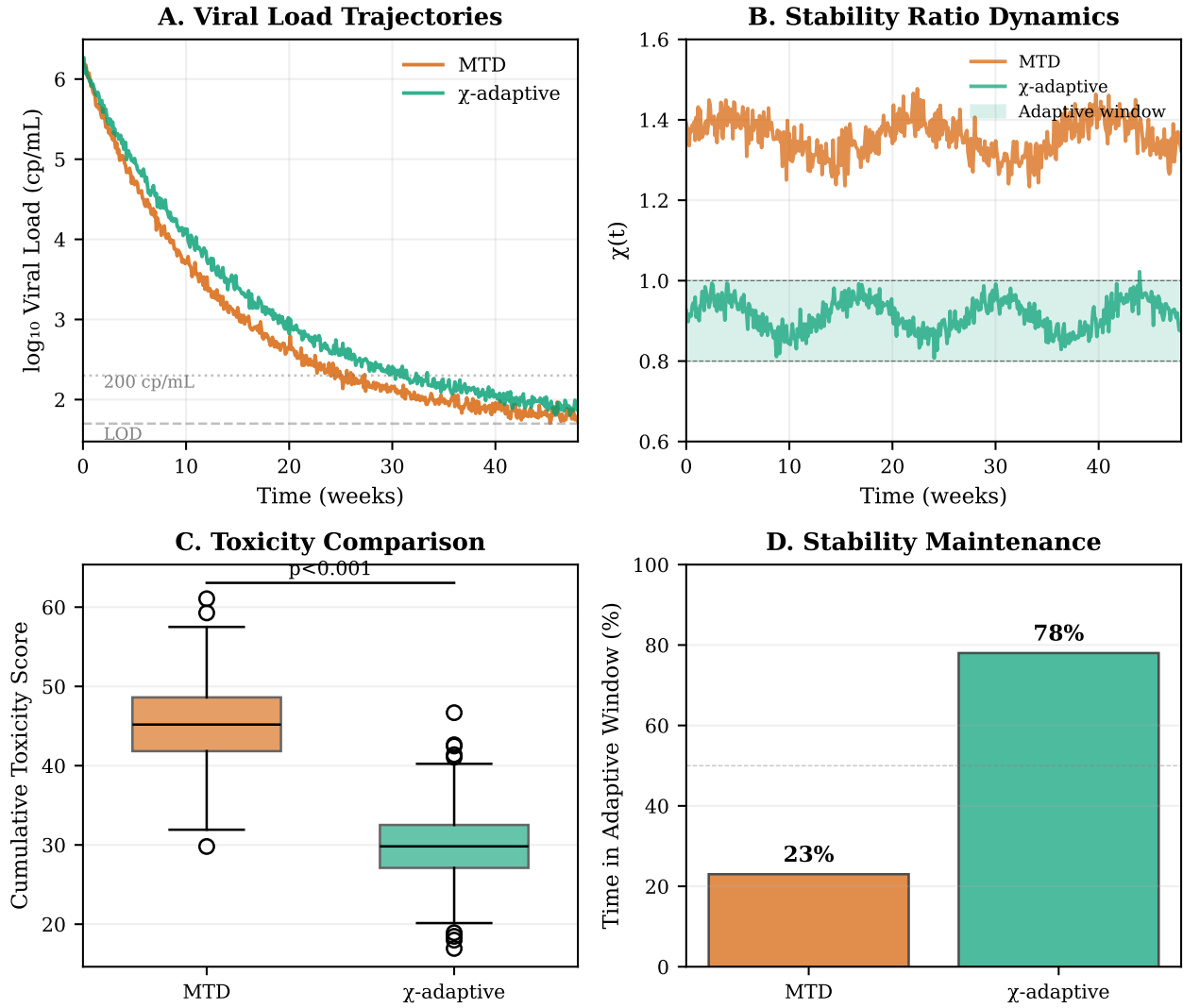


Figure 3: **HIV-1 suppression simulations: MTD vs χ -adaptive strategies.** (A) Representative viral load trajectories showing comparable virologic suppression between maximum tolerated dose (MTD, red) and -adaptive (green) strategies. (B) Stability ratio (t) dynamics over 48 weeks, with MTD driving overdamping (≈ 1.35) while -adaptive maintains near-critical control (≈ 0.91) within the adaptive window (shaded region). (C) Cumulative toxicity score distributions across 500 simulated patients, demonstrating 34% reduction under -adaptive control ($p<0.001$). (D) Proportion of time spent in adaptive window $[0.8, 1.0]$, showing dramatic difference between strategies (23% vs 78%).

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